

COURSE NAME

SOFTWARE QUALITY
AND TESTING

CSC 4133

(UNDERGRADUATE)

CHAPTER 01

BASICS OF SOFTWARE QUALITY ASSURANCE

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WHAT IS A SOFTWARE?

- ❑ Software is:
 - More than just a computer program!
 - Composed of computer programs, procedures, and possibly associated documentation and data related to the operation of a computer system.
- ❑ Two major types of Software:
 - **Generic (Buy)** – Stand alone, Sold on open market (OTH)
 - **Customized (Build)** – For specific customer or business
 - **BUY & BUILD** – Most software that you buy for personal use will likely be ‘commercial off-the-shelf’ (COTS), means the same application will be run by thousands of other users. It may not work exactly how you’d like it to or do everything you’d want it to whereas a customized (bespoke) software solution is built exactly to the business’ specific requirements.



GENERIC VS. CUSTOMIZE SOFTWARE

- ❑ **Expensive:** Customize software is more expensive than Generic software
- ❑ **Copyright:** Customer holds the copyright in Customized software where Software Development Organization holds the copyright in Generic software
- ❑ **Competitive advantages:** Customize software can provide software's competitive advantages
- ❑ **Functional use:** Generic software can have unused and/or missing functions
- ❑ **Performance (Less resource consumption):** Performance of customize software can be better than Generic software.
- ❑ **Reliability:** Generic software is more reliable than customize software

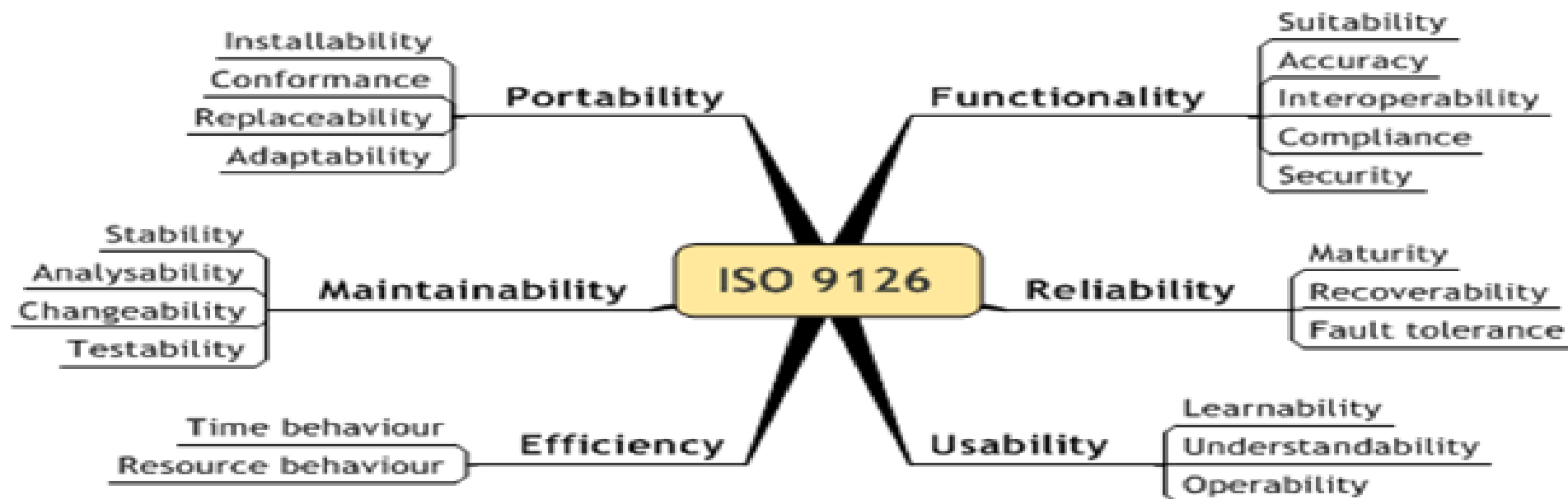
WHAT IS SOFTWARE QUALITY?

❑ Software Quality (as per ISO/ IEC 9126):

The totality of functionality and features of a software product that contributes to its ability to **satisfy** stated or implied **needs**. It can be customized for organizations.

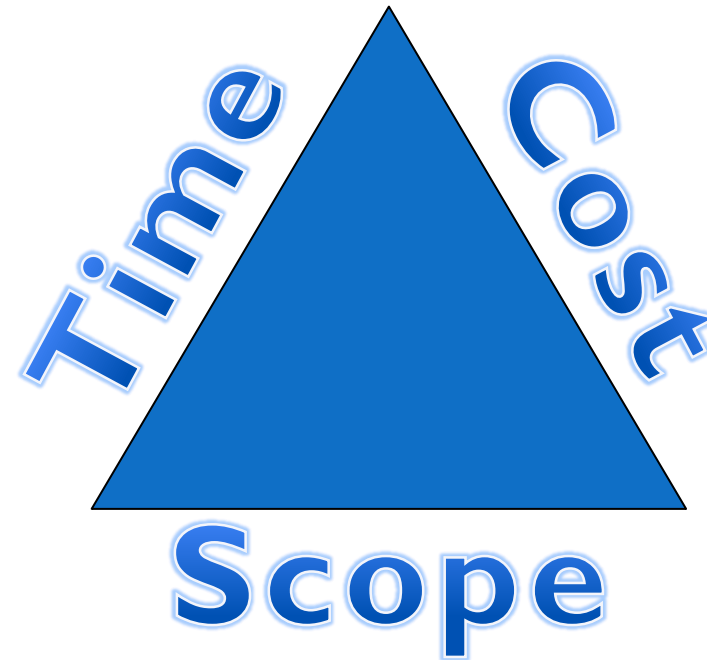
❑ Software Quality (as IEEE Std 610):

The degree to which a component, system or process **meets specified requirements** and/or **user/customer needs and expectations**.



CHALLENGES IN SOFTWARE PROJECTS

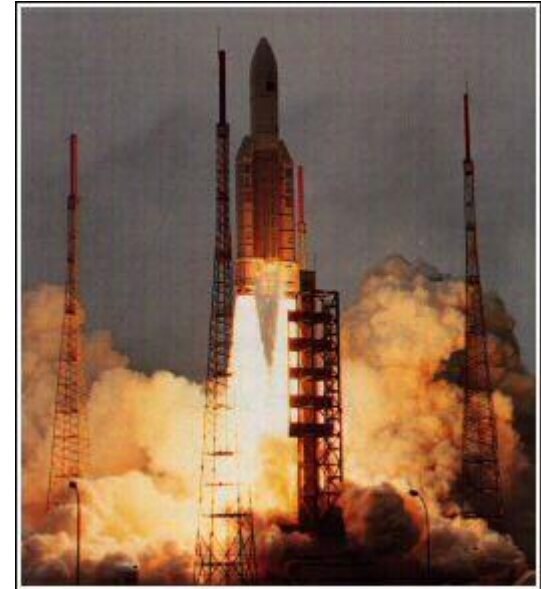
- ❑ The primary goal of project management activity is to deliver the product on **time**, on **budget** and expected **quality**
- ❑ What's the main challenges of software development now-a-days?
 - **Time**: late, difficult to deliver on time
 - **Cost**: high cost, over budget
 - **Scope**: low quality with faults



EXAMPLE OF SOFTWARE DEFECTS

❑ Flight Ariane 5

- Most Expensive Computer Bug in History
- On June 4, 1996, the rocket Ariane 5 tore itself apart 37 seconds after launch because of a malfunction in the control software making the fault most expensive computer bug in history.



❑ Lethal X-Rays :Therac-25 system

Therac-25 was a radiation therapy machine produced by Atomic Energy of Canada Ltd (AECL) in 1986. But initially lot of people died because of massive overdose of radiation. And this is happened because of a software bug.

SOFTWARE TESTING

❑ Software Testing is the process of executing a system or component under specified conditions with the intent of **finding defects/bugs** and to verify that it **satisfies specified requirements**.

❑ Goals of Software Testing

- Main goal of testing is to **detect issues**
 - *Firmness*: Errors in system operation
 - *Commodity*: Requirement conformance
 - *Delight*: System performance (TCAS in Autopilot system)
- Have different levels of testing operation
 - Unit testing in developing each module
 - Integration testing to combine different module and check incompatibility
 - System testing is done at system combination of software and hardware



ROLE OF TESTING

Software quality assessment divide into two categories:

❑ Static analysis (manual testing)

- It examines the code/document and reasons over all behaviours that might arise during run time
- **Examples:** code review, inspection, and algorithm analysis
- **Limitation:** takes longer time in review process

❑ Dynamic analysis (automated testing)

- Actual program execution to expose possible program failure
- One observe some representative program behaviour, and reach conclusion about the quality of the system
- **Limitation:** It can not think out of the box (supervised learning) finds only syntax error

❑ Static and Dynamic Analysis are complementary in nature. Focus is to combine the strengths of both approaches



Static Testing



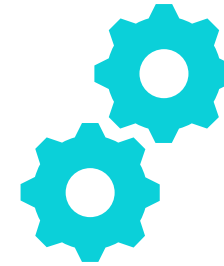
Dynamic Testing

VALIDATION AND VERIFICATION



Validation

- Software systems must do what they are supposed to do; they must **do the right things**
- Validation is a process that ensures the software product meets the customer requirements.
- Building the correct product

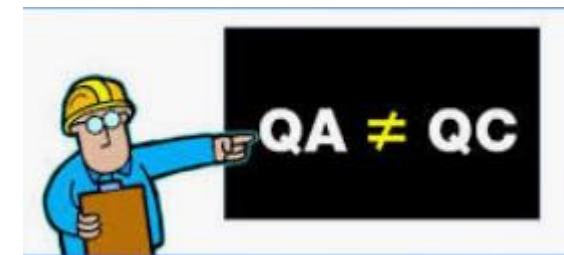
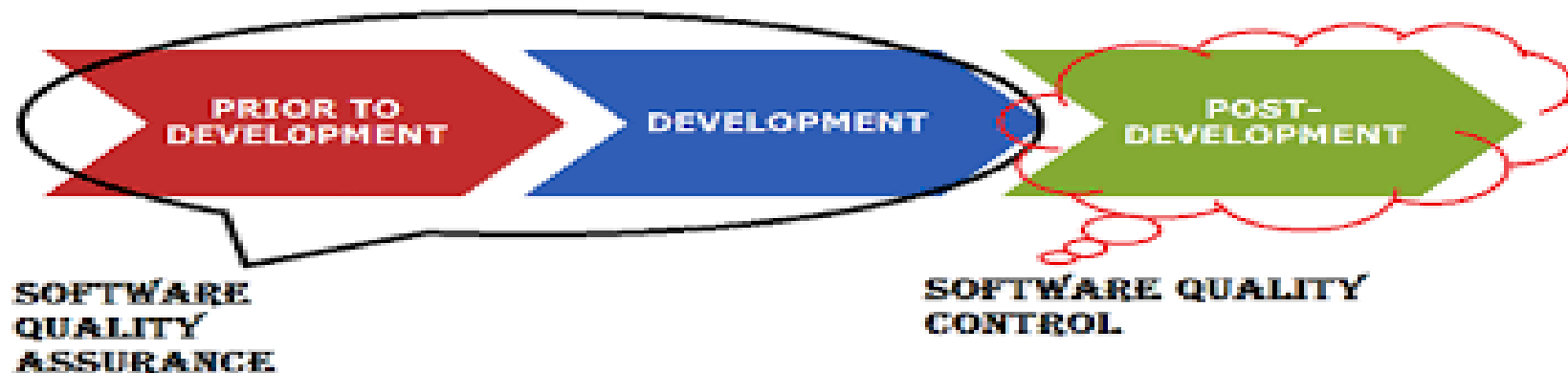


Verification

- Software systems must perform the specific tasks correctly; they must **do the things right**
- Verification is a process that ensures the software product has been developed properly.
- Building the product correctly

DIFFERENCE BETWEEN SQA & SQC

SQA (Software Quality Assurance)	SQC (Software Quality Control)
A process which deliberates on providing assurance that quality request will be achieved (Planning for doing a process)	A process which deliberates on fulfilling the quality request (Action for executing the planned process)
Is the technique of managing the quality	Is the method to verify the quality
Does not involve executing the program	Always involves executing the program
All team members are responsible	Only testing team is responsible



DIFFERENCE BETWEEN SQA & SQC

SQA	SQC
Verification - Makes sure you are doing the things right (is process oriented)	Validation - Makes sure the results of what you've done are what you expected (product oriented)
Responsible for full software development life cycle (SDLC)	Responsible for software testing life cycle
Detects weakness. Aim is to prevent the defect	Detects defects. Aim is to identify and resolve the defects
Is failure prevention system	Is failure detection system

DIFFICULTIES ON QUALITY EXPECTATIONS

❑ Difficulties in achieving good quality:

- **Size (LOC in the function code)**
Manual testing would be difficult to find logical errors in bulky source code
- **Invariant Complexity in the Product and Project**
Complexity in S/W is more because it deals with logic where **any error** in logic may cause significant impact such as **autopilot** software (e.g. set landing mode)
- **Environmental stress/constraints**
Test the system with it maximum load (e.g. SSC/HSC result publishing website)
- **Flexibility/adaptability expected**
Frequent change because of flexibility in S/W (e.g. regression testing after each change)
- **Cost and market conditions**
Testing cost with automated tools and license

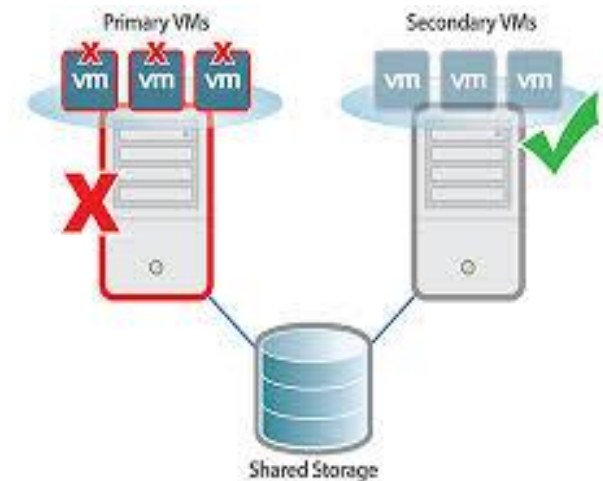


SOFTWARE QUALITY ENGINEERING (SQE)

- ❑ To ensure software quality through the related **validation** and **verification** activities.
- ❑ To be **carried** out by the **people and organizations** responsible for developing and supporting these software systems in an overall quality engineering process:
 - **Planning:** Quality planning
 - **Execution:** Execution of selected QA plan or software validation & verification activities
 - **Decision:** Decide pass/fail criteria of a test case. Measurement and Analysis to provide convincing evidence to demonstrate software quality to all parties involved
- ❑ Customers and users need to have the assurance that their quality expectations are satisfied by the delivered software systems.

SQE ACTIVITIES

- ❑ **Testing:** remove defect & ensure quality
- ❑ Other QA alternatives to testing
 - Defect Prevention
(plan for avoid any defect)
 - Formal Verification
(e.g. Inspection/Review/Walkthrough)
 - Fault Tolerance
(e.g. data backup, autosave document, system reliability up to 5 failure)



ERROR, FAULT, FAILURE, AND DEFECT

Error

- A human action that produces an incorrect result
- Missing or incorrect human action resulting in certain fault(s) being injected into a software

Fault

- An incorrect step, process, or data definition in a computer program
- An underlying condition within a software that causes certain failure(s) to occur

Failure

- The inability of a system or component to perform its required functions within specified performance requirements
- A behavioural deviation from the user requirement or the product specification

Defect

- Failures, faults, and errors are collectively referred to as defects in literature.

Note: Software problems or defects, are also commonly referred to as “bugs”

COMPLETE TESTING

❑ The objectives of Testing is to Detect bugs

- Reduce the risk of defects
- Reduce the cost of testing

❑ Complete/Exhaustive testing

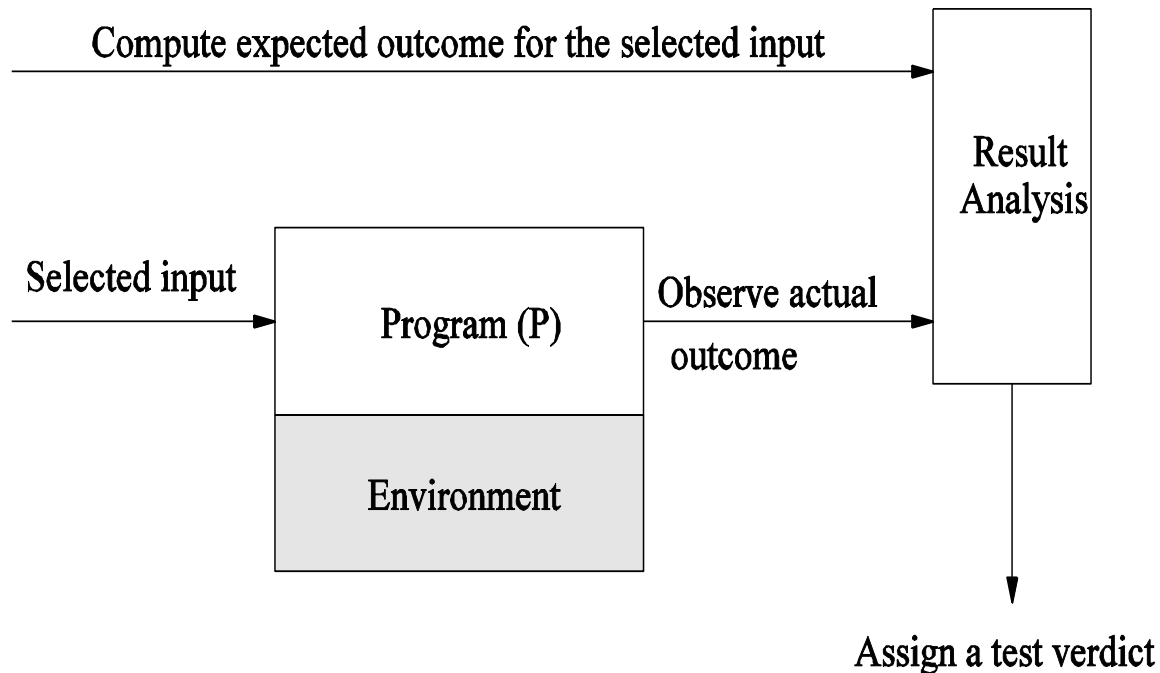
- There are no undisclosed faults at the end of test phase

❑ Complete testing is nearly impossible for most of the systems, Because..

- The domain of possible inputs of a program is too large (valid inputs and invalid inputs)
- The design issues may be **too** complex to completely test (testing effort may very high for testing the system completely and for manually test a system completely)
- It may **not be possible** to create all possible execution environments of the system (create all possible simulation of an **auto pilot testing**)

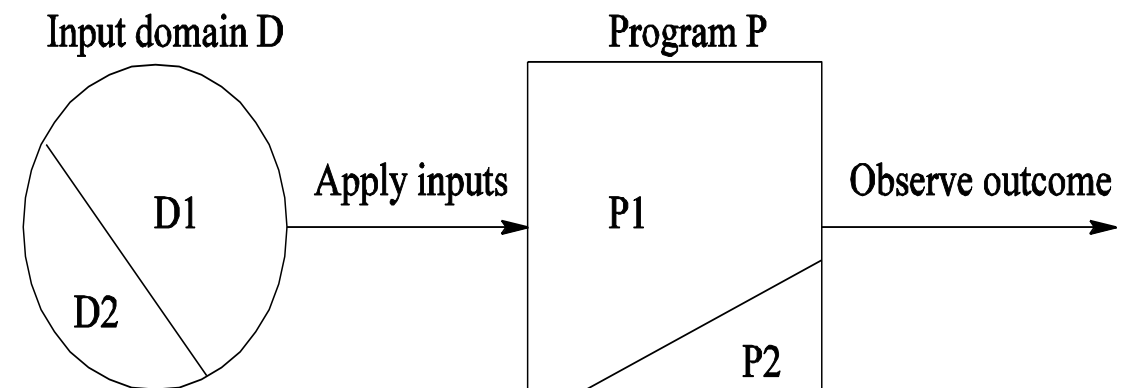


TESTING ACTIVITIES



- Identify the objective to be tested (prioritized)
- Select inputs from input domain
- Compute the expected outcome
- Set up the execution environment of the program
- Execute the program
- Analyse the test results with expected outcome

- A subset of the input domain exercising a subset of the program behaviour
- Divide the input domain D into $D1$ and $D2$
- Select a subset $D1$ of D to test program P
- It is possible that $D1$ exercise only a part $P1$ of P



REFERENCES

- ❑ Software Testing And Quality Assurance – Theory and Practice - Kshirasagar Naik & Priyadarshi Tripathy
- ❑ Software Quality Engineering: Testing, Quality Assurance and Quantifiable Improvement - Jeff Tian